

APPLICATION DEVELOPMENT

Software Requirements Specification (SRS) **Smart Farming and Precision Agriculture Management System**

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Software Requirements Specification (SRS)

Smart Farming and Precision Agriculture Management System

1. Introduction

1.1 Purpose

Defines requirements for a Smart Farming Platform enabling farmers to manage crop planning, IoT sensor monitoring, precision inputs, pest alerts, and yield analytics.

1.2 Scope

Enables farmers and agricultural officers to manage farm registration, crop planning, IoT sensor data, precision irrigation, fertiliser scheduling, and yield tracking. Key functionalities:

- Farmer and agricultural officer authentication
- Farm registration with GIS parcel mapping
- Crop season planning with variety selection
- IoT soil and weather sensor integration
- Precision irrigation scheduling with alerts
- Fertiliser and pesticide recommendation engine
- Pest and disease alert from image analysis
- Drone imagery integration for crop health
- Market price and mandi integration
- Analytics for yield, input cost, and profitability

1.3 Definitions, Acronyms, and Abbreviations

- IoT: Internet of Things
- GPS: Global Positioning System
- GIS: Geographic Information System
- NPK: Nitrogen Phosphorus Potassium
- API: Application Programming Interface

1.4 References

- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications
- Spring Boot Security Documentation
- React.js Documentation
- Precision Agriculture Best Practices
- ICAR Crop Management Guidelines

1.5 Overview

Covers all requirements for the Smart Farming and Precision Agriculture Management System with role-based security and comprehensive smart farming management.

2. Overall Description

2.1 Product Perspective

Spring Boot + React.js + JWT + JPA for smart farming operations.

2.2 Product Functions

- User registration and authentication with smart farming role-based access
- Complete CRUD operations for all smart farming records
- Farm lifecycle management from creation through completion
- Real-time status tracking and milestone notifications
- Smart Farming analytics and performance reporting
- Document management with version control
- Integration with relevant external systems
- Mobile-responsive access for all user types
- Notification system for all key events
- Compliance reporting and audit trail management

2.3 User Classes and Characteristics

- System Administrator: Full smart farming system access and configuration
- Smart Farming Manager: Comprehensive operational management
- Smart Farming Supervisor: Day-to-day workflow management
- Smart Farming Officer: Operational task execution
- Farm: Own record access and self-service functions
- Guest: Read-only access to public information

2.4 Operating Environment

- Server: Spring Boot 3.x with Spring Security (Java 17+)
- Database: JPA-compatible relational database with smart farming management schema
- Frontend: React.js with authentication components (port 8081)
- Backend: Spring Boot with Spring Security REST APIs (port 8080)
- Authentication: JWT-based with role-specific token claims and expiry
- Testing: JUnit 5 for backend, React Testing Library for frontend
- Deployment: Docker containerised on cloud (AWS, Azure, or GCP)

2.5 Design and Implementation Constraints

- Spring Security and JWT on all smart farming endpoints
- Name: alphabetic only; Phone: exactly 10 digits
- All farm records require validation before saving
- Role-based access enforced at both React route and API level
- All data modifications logged with user identity and timestamp

2.6 User Documentation

- Smart Farming manager user guide
- Farm self-service portal guide
- Administrator configuration guide
- Compliance and reporting guide

2.7 Assumptions and Dependencies

- Users have internet access for smart farming system access
- Integration with relevant smart farming external systems available
- Email and SMS services available for notifications
- Cloud infrastructure available for deployment

3. Specific Requirements

3.1 External Interfaces

3.1.1 User Interfaces

- Landing Page: Smart Farming and Precision Agriculture Management System overview with login
- Farm Dashboard: Core smart farming management interface
- Analytics: Smart Farming performance charts
- Admin Panel: User management and configuration

3.1.2 Hardware Interfaces

- Document scanner
- Digital signature pad
- Barcode/QR scanner

3.1.3 Software Interfaces

- JWT REST API
- Email service
- SMS gateway
- Document storage

3.1.4 Communication Interfaces

- HTTPS with TLS 1.3 for all client-server communications
- JSON with security headers (CORS, CSP, X-Frame-Options) on all responses
- JWT HMAC-SHA256 signature verification on every API request
- WebSocket for real-time notifications and live dashboard updates

3.2 Functional Requirements

User Authentication and Account Management

FR1: User Registration and Authentication

Description: Provide smart farming-specific registration.

Detailed Requirements:

- Registration captures name, user ID, email, role, smart farming assignment, and contact
- Email OTP verification and ID validation required before activation
- Multi-role support handles Guest, Farmer, Field Officer, Agricultural Officer accounts
- Supervisor approval required for all operational roles before platform access
- Password enforces minimum 8 characters with uppercase, lowercase, digit, and special character
- Account recovery uses time-limited email OTP with 15-minute validity window
- Profile management allows post-registration updates with complete audit trail
- Security logging records every event with IP address and timestamp
- Two-factor authentication mandatory for admin and financial operations
- All smart farming operational data requires authentication — no anonymous access

FR2: Session Management

Description: Secure sessions for smart farming operations.

Detailed Requirements:

- JWT tokens with role-specific expiry durations
- Multi-credential login: user ID, email, or mobile
- Auto-logout after 30 minutes of inactivity
- Device recognition with extra verification for new devices
- Progressive lockout: 15, 30, 1440 minutes after 5, 10, 15 failures
- SSO with corporate systems
- All events logged for compliance audit

- Token blacklisting on logout and breach events
- Emergency override for critical operational scenarios
- Full session audit trail maintained

FR3: Role-Based Access Control

Description: Granular RBAC for smart farming data.

Detailed Requirements:

- Users access only own records; managers access assigned team scope; admin accesses everything
- Hierarchical configurable permissions per role and function
- Temporal controls enforce operational hours
- Sensitive operations restricted by function permissions
- Users see only data relevant to their role and assignment scope
- Every access logged with user identity, timestamp, and declared purpose
- Role changes take effect on the user's next login
- Cross-department access requires approval
- Guest access limited to public information only
- Permission matrix maintained in database for runtime enforcement without code deployment

Core System Operations

FR4: Farm Registration and Management

Description: Register and manage farm records.

Detailed Requirements:

- Registration form captures all required smart farming fields with complete format validation
- Document upload and verification workflow with expiry date tracking
- Unique ID generation following configured structured format
- Status lifecycle with all transitions logged
- Bulk CSV import with field-level validation and detailed error reporting
- Version history maintained for all record modifications
- Search and filter across all record types by multiple criteria
- Duplicate detection on unique identifiers before record creation
- External verification system integration where applicable
- Registration volume and status distribution analytics dashboard

FR5: Workflow Processing

Description: Process smart farming workflows end-to-end.

Detailed Requirements:

- Workflow captures all required smart farming operational fields with validation
- Multi-step approval process with configurable workflow stages
- Status update notifications dispatched at each workflow stage transition
- SLA tracking per workflow type with breach alerts and automatic escalation
- Document attachment supported at each workflow stage for evidence
- Complete history log with full chronological audit trail
- Bulk workflow operations for operational efficiency
- Integration with related external systems and data sources
- Workflow analytics: average cycle time and completion rate by type

- Exception handling with documented override procedures and approval

FR6: Scheduling and Resource Management

Description: Manage smart farming scheduling and resource allocation.

Detailed Requirements:

- Schedule creation with resource assignment and time slot configuration
- Availability check performed before schedule confirmation
- Notification dispatched to all participants on schedule confirmation
- Calendar view with filter and search across all schedule types
- Rescheduling workflow with mandatory reason capture and notification
- Cancellation workflow with all stakeholder notification
- Conflict detection prevents double-assignment of resources
- Reminder notifications at configured intervals before scheduled events
- Utilisation analytics for capacity planning and optimisation
- Recurring schedule configuration for regular operational events

FR7: Document and Compliance Management

Description: Manage smart farming documents and regulatory compliance.

Detailed Requirements:

- Document upload with metadata, category, and expiry date capture
- Version control with full rollback capability to any previous version
- Access control configured per document type and sensitivity level
- Expiry alerts dispatched at configured days before document due date
- Digital signature capture for approvals and authorisations
- Complete audit trail for all document access and modification events
- Bulk document operations for migration and management efficiency
- Full-text search across the document repository by keyword and metadata
- Retention policy with automated archival triggered on retention date
- Export in PDF and other standard formats for submission and reporting

FR8: Communication and Notification System

Description: Manage smart farming stakeholder communications.

Detailed Requirements:

- Multi-channel notifications: in-app push, email, and SMS per user preference
- Event-triggered automated notifications on all key system events
- Configurable notification templates with variable substitution
- Delivery status tracking with automatic retry for failed deliveries
- Broadcast capability to defined stakeholder groups by role or segment
- Reminder sequence for pending actions at configured intervals
- Digest emails for activity summaries to managers and administrators
- Escalation alerts for SLA breaches dispatched to supervisors
- User notification preference management per channel and type
- Communication audit trail maintained for all notifications

FR9: Analytics and Reporting Dashboard

Description: Comprehensive analytics and reporting for smart farming.

Detailed Requirements:

- Real-time smart farming operational KPI dashboard with drill-down capability
- Trend charts showing daily, weekly, and monthly performance over time
- Custom parametric reports with date range and dimension filters
- Comparison reports against configured targets and historical benchmarks
- Export in PDF, Excel, and CSV formats for distribution and filing
- Automated scheduled reports distributed to configured stakeholder list
- Drill-down capability from summary view to individual transaction level
- Predictive analytics from historical patterns for forward planning
- Role-specific dashboards showing metrics most relevant to each role
- Mobile-accessible dashboard for management review on any device

FR10: Integration and External Connectivity

Description: External system integration for smart farming.

Detailed Requirements:

- REST API integration with all primary external operational systems
- Webhook support for real-time event broadcasting to connected systems
- File-based integration via validated CSV or XML for batch operations
- OAuth 2.0 or API key authentication for all external connections
- Error handling with retry logic and dead-letter queue for failures
- Integration monitoring dashboard displaying connection status and errors
- Data transformation layer for field and format mapping between systems
- Bulk synchronisation support for initial data migration workloads
- Staging integration environment for development and testing
- SLA monitoring for all integration health metrics

FR11: Mobile and Remote Access

Description: Mobile access for smart farming field operations.

Detailed Requirements:

- Mobile-responsive web application optimised for smartphone use
- Native app with offline capability for poor-connectivity environments
- Push notifications for all real-time operational alerts
- Camera integration for photo capture and document scanning
- GPS location services for geo-tagged operational records
- Biometric or PIN security enforced on all mobile devices
- Background synchronisation on connectivity restoration
- Touch-optimised UI with large interactive elements for ease of use
- Offline mode indicator clearly displaying cached content and last sync time
- Mobile analytics tracking feature adoption and session patterns

FR12: Security and Compliance

Description: Security and compliance management for smart farming.

Detailed Requirements:

- AES-256 at rest for all sensitive data
- TLS 1.3 for all communications
- Role-based access prevents any cross-user data leakage at API level
- Immutable audit logging for all data access and modification events
- Monthly automated vulnerability scans plus quarterly internal review
- Data retention and archival policies automatically enforced

- Encrypted backups stored in geographically separate data centres
- Annual professional penetration testing
- GDPR-aligned data deletion procedures
- On-demand compliance audit packages for regulatory inspection readiness

FR13: Advanced Analytics and Predictive Intelligence

Description: Predictive analytics for smart farming.

Detailed Requirements:

- Predictive demand forecasting using smart farming historical operational data
- Risk scoring for prioritised management attention and intervention
- Anomaly detection with automated alerts for unusual patterns
- Comparative benchmarking against peer organisations and industry standards
- Optimisation recommendations generated from operational performance data
- Cohort analysis for trend identification and pattern recognition
- Machine learning model for outcome prediction and planning support
- Visual analytics including heat maps, trend lines, and funnel diagrams
- Natural language weekly insight summary for management consumption
- Evidence-based strategic planning tools from consolidated analytics

System Administration and Security

FR14: Exception Handling

Description: Comprehensive exception handling for smart farming.

Detailed Requirements:

- Name validation: 'Name must not contain special characters or numbers'
- Phone validation: 'Phone Number must be exactly 10 digits long'
- DuplicateFarmException prevents duplicate unique identifier creation
- Custom exceptions in com.examly.springapp.exception: InvalidNameException, InvalidPhoneException, DuplicateFarmException
- Structured logs capture exception type, user context, affected record, and timestamp
- User-facing messages are clear and actionable without exposing system internals
- Exponential back-off retry for all transient external service failures
- Graceful degradation on database and external API failures
- Exception analytics track frequency by type and mean resolution time
- Integration errors logged and escalated to operations team immediately

FR15: Data Security

Description: Comprehensive data protection for smart farming.

Detailed Requirements:

- AES-256 encryption at rest for all sensitive data fields
- TLS 1.3 plus HSTS in transit
- RBAC enforced at every API service layer endpoint
- Server-side sanitisation against SQL injection and XSS
- Immutable audit logging of all data access and modification events
- Real-time intrusion detection with automated blocking
- Automated archival and deletion policies enforced
- Encrypted backups stored in geographically separate data centres

- Monthly automated scans and annual professional penetration testing
- Compliance audit packages generated on regulatory request

FR16: System Administration

Description: Comprehensive administrative tools for lifecycle and compliance.

Detailed Requirements:

- User lifecycle: creation, role changes, suspension, deactivation with full audit
- System configuration updates through admin portal without code redeployment
- Real-time admin dashboard with users, activity, and system health metrics
- Bulk CSV import with field validation and detailed error reporting
- System health monitoring: API response, error rates, and storage utilisation
- Daily encrypted backups with weekly restoration testing
- Searchable tamper-evident audit trail
- Integration monitoring dashboard for all external API connections
- Notification template management for all system communications
- Compliance calendar tracking all regulatory deadlines and licence renewals

FR17: Advanced Analytics

Description: Analytics and predictive intelligence for smart farming.

Detailed Requirements:

- Real-time smart farming operational KPI dashboard with trend charts
- Custom parametric report builder with date range and dimension filters
- Data export in PDF, Excel, and CSV for management and regulatory reporting
- ML-powered anomaly detection identifying unusual patterns requiring attention
- Predictive modelling for demand and resource requirement forecasting
- Comparative benchmarking against industry or peer-group performance standards
- User behaviour analytics identifying workflow bottlenecks and optimisation opportunities
- Automated weekly insight reports for configured management stakeholders
- Risk scoring model ranks records by composite risk factors for prioritised intervention
- Evidence-based strategic planning recommendations generated from analytics data

3.3 Performance Requirements

- Authentication: Login < 1 second for 95% of requests
- Record management: Create, read, update, delete < 2 seconds for 90% of requests
- Search: Results returned < 1.5 seconds for complex multi-field queries
- Validation: Combined client and server validation < 1 second for 95% of submissions
- Concurrent users: 500+ simultaneous users without performance degradation
- Reports: Standard reports generate in < 10 seconds for typical data volumes
- Notifications: End-to-end delivery within 5 seconds of trigger event

3.4 Logical Database Requirements

Core Tables:

- Users Table: id, name, phone_number, email, password_hash, role, created_date, last_login, is_active
- Farms Table: id, farmer_id, farm_name, district, total_area_acres, soil_type, lat, lng, status
- CropSeasons Table: id, farm_id, crop_name, variety, sowing_date, expected_harvest, area_acres, status, actual_yield_kg
- SensorReadings Table: id, farm_id, sensor_type, value, unit, recorded_at, alert_triggered

- InputRecommendations Table: id, crop_season_id, recommendation_type, recommended_quantity, recommended_date, actual_applied_date, status

3.5 Design Constraints

- Spring Security+JWT on all endpoints
- React protected routes + mobile-responsive
- JPA transactions for data integrity
- All changes logged with user and timestamp
- Name: alphabetic only; Phone: exactly 10 digits
- RBAC at React route and Spring Boot API level

3.6 Software System Attributes

3.6.1 Reliability

- 99.9% uptime with planned maintenance communicated 48 hours ahead
- Graceful non-sensitive error messages for all system failures
- Multi-layer data integrity validation at client, API, service, and database levels
- Automatic failover for critical smart farming operations within 60 seconds
- Daily encrypted backups with quarterly restoration testing

3.6.2 Availability

- 24/7 availability with maintenance outside peak operational hours
- Load balancing across minimum two application server instances
- Geographic data replication to secondary data centre for disaster recovery
- Real-time health monitoring with automated on-call escalation
- Horizontal auto-scaling for peak load periods

3.6.3 Security

- Authentication: JWT HS256 with role-specific session duration configuration
- Authorisation: Role-based access control at API service layer for every endpoint
- Data Protection: AES-256 at rest and TLS 1.3 in transit for all sensitive data
- Input Validation: Server-side sanitisation against injection and XSS attacks
- Audit Trail: Immutable logging of all data access and modification events
- Compliance: Regulatory standards verified and documented at every major release

3.6.4 Maintainability

- Layered architecture: Controller, Service, Repository, Domain Model layers
- Minimum 85% unit and integration test coverage measured in CI pipeline
- Swagger/OpenAPI 3.0 documentation with examples for every endpoint
- Git feature branch workflow with mandatory peer code review
- Spring Boot profiles with encrypted secret management

3.6.5 Portability

- Docker + Docker Compose for consistent local and production deployments
- Spring Data JPA: MySQL 8.0+ and PostgreSQL 14+ without code changes
- Cloud manifests: AWS ECS, Azure AKS, GCP GKE
- Spring Boot profiles: dev, test, staging, production
- Cross-browser: Chrome 100+, Firefox 100+, Edge 100+, Safari 15+

3.7 Other Requirements

- Testing: Backend 85% JUnit 5 and Mockito; Frontend 75% React Testing Library
- Security Testing: Quarterly OWASP ZAP + annual professional penetration testing
- Performance Testing: Monthly JMeter load tests before each major release
- Documentation: API reference, deployment runbook, user manuals each release

- Compliance: Agriculture regulatory standards verified at every major release
- Accessibility: WCAG 2.1 Level AA for all user-facing interfaces

4. Appendices

Appendix A: Role-Based Permission Matrix

Functionality	Guest	Farmer	Field Officer	Agricultural Officer	Extension Officer	Admin
View Agri Info	✓	✓	✓	✓	✓	✓
Register Farm	-	✓	L	✓	✓	✓
View Own Farm	-	✓	✓	✓	✓	✓
Plan Crop Season	-	✓	✓	✓	✓	✓
View Sensor Data	-	✓	✓	✓	✓	✓
Get Recommendations	-	✓	✓	✓	✓	✓
Report Pest/Disease	-	✓	✓	✓	✓	✓
View Yield Analytics	-	✓	L	✓	✓	✓
Manage Farmers	-	-	✓	✓	✓	✓
Manage Users	-	-	-	-	-	✓

Appendix B: Database Schema Details

Users Table:

- id (PRIMARY KEY)
- name (VARCHAR, NOT NULL) — alphabetic only
- phone_number (VARCHAR 10, NOT NULL) — exactly 10 digits
- email (VARCHAR, UNIQUE, NOT NULL)
- password_hash (VARCHAR, NOT NULL)
- role (ENUM)
- created_date (TIMESTAMP)
- last_login (TIMESTAMP)
- is_active (BOOLEAN DEFAULT TRUE)

Farms Table:

- id (PRIMARY KEY)
- farmer_id (FK)
- farm_name (VARCHAR)
- district (VARCHAR)
- total_area_acres (DECIMAL)
- soil_type (VARCHAR)
- lat (DECIMAL)
- lng (DECIMAL)
- status (ENUM: ACTIVE,FALLOW,LEASED)

CropSeasons Table:

- id (PRIMARY KEY)
- farm_id (FK)
- crop_name (VARCHAR)
- variety (VARCHAR)
- sowing_date (DATE)
- expected_harvest (DATE)
- area_acres (DECIMAL)
- status (ENUM: PLANNED,SOWING,GROWING,HARVESTED,FAILED)
- actual_yield_kg (DECIMAL)

SensorReadings Table:

- id (PRIMARY KEY)
- farm_id (FK)
- sensor_type (ENUM: SOIL_MOISTURE,TEMPERATURE,HUMIDITY,PH,NPK)
- value (DECIMAL)
- unit (VARCHAR)
- recorded_at (TIMESTAMP)
- alert_triggered (BOOLEAN DEFAULT FALSE)

InputRecommendations Table:

- id (PRIMARY KEY)
- crop_season_id (FK)
- recommendation_type (ENUM: IRRIGATION,FERTILISER,PESTICIDE)
- recommended_quantity (VARCHAR)
- recommended_date (DATE)
- actual_applied_date (DATE)
- status (ENUM: PENDING,APPLIED,SKIPPED)

Appendix C: Validation Rules Summary

Field	Validation Rules	Error Messages
Name	Required, alphabetic characters and spaces only, 2–100 characters	"Name is required" / "Name must not contain numbers or special characters"
Phone Number	Required, exactly 10 consecutive numeric digits	"Phone Number is required" / "Phone Number must be exactly 10 digits long"
Email	Required, valid email format, unique per account	"Please enter a valid email address" / "This email is already registered"
Password	Minimum 8 characters with uppercase, lowercase, digit, and special character	"Password must meet security requirements"
Farm Area	Positive decimal in acres	"Area must be a positive number"

Sensor Value	Within physiological range for type	"Sensor value is outside valid range"
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Appendix D: Security Implementation Details

JWT Configuration:

- Token Expiration: 8 hours for regular users, 12 hours for staff, 24 hours for admins
- Refresh Token: 30-day validity with automatic rotation on each use
- Algorithm: HS256 with minimum 256-bit secret key stored as environment variable
- Secret Key: Never hardcoded in source code — loaded from secure environment config
- Token Blacklisting: Implemented for logout, password change, and security breaches
- Token Claims: Include userId, role, email, issuedAt, and expiration timestamp

Password Policy:

- Minimum 8 characters with uppercase, lowercase, digit, and special character
- Cannot match last 5 passwords; expires 90 days staff / 180 days users
- Lockout 15 minutes after 5 consecutive failed login attempts
- Secure password reset via email OTP with 15-minute validity window
- Password strength indicator displayed in real time during registration

Data Security:

- Sensitive smart farming data encrypted at rest using AES-256
- All communications use HTTPS with TLS 1.3 and HSTS headers
- Server-side input sanitisation preventing SQL injection and XSS
- Role-scoped data access controls enforced at service layer
- Tamper-evident encrypted audit logs for all security events

Rate Limiting:

- Login: maximum 5 attempts per 15-minute window per IP address
- Registration: maximum 3 new accounts per day per IP
- Search: maximum 1,000 queries per hour per authenticated user
- Data writes: maximum 100 operations per hour per user
- Password reset: maximum 3 emails per hour per address

Appendix E: Exception Handling Classes

The following custom exception classes shall be implemented in `com.examly.springapp.exception`:

InvalidNameException

Thrown when a smart farming name field contains non-alphabetic characters or digits.

- Extends: `RuntimeException`
- Trigger: Non-alphabetic characters in name field
- HTTP Status: 400 Bad Request

InvalidPhoneException

Thrown when a phone number is not exactly 10 consecutive digits.

- Extends: `RuntimeException`
- Trigger: Phone not exactly 10 digits
- HTTP Status: 400 Bad Request

DuplicateFarmException

Thrown when a smart farming record with the same unique identifier already exists.

- Extends: RuntimeException
- Trigger: Duplicate unique identifier on create
- HTTP Status: 409 Conflict

UnauthorisedAccessException

Thrown when a user accesses a resource outside their assigned role.

- Extends: RuntimeException
- Trigger: API call beyond user role permissions
- HTTP Status: 403 Forbidden

ResourceNotFoundException

Thrown when a smart farming record cannot be found by the provided ID.

- Extends: RuntimeException
- Trigger: GET/PUT/DELETE with non-existent ID
- HTTP Status: 404 Not Found

Appendix F: Business Rules and Constraints

Farm Rules:

- All mandatory farm fields must pass validation before submission
- Name must contain alphabetic characters and spaces only — no digits or special characters
- Phone number must be exactly 10 consecutive numeric digits
- Duplicate farm detection prevents creation of records with the same unique identifier
- Status changes are logged with the responsible user identity and timestamp

Data Retention:

- Operational records retained per applicable regulatory requirement
- Audit logs retained for 7 years minimum
- Archived records accessible in read-only mode
- Data deletion procedures documented and verified annually

Appendix G: Future Enhancement Roadmap

Phase 2 Features:

- Native iOS and Android apps with offline smart farming capability
- Advanced analytics with machine learning trend forecasting
- Third-party platform integration via REST and webhook APIs
- Automated recommendation algorithms reducing manual work
- Multi-language support for diverse user populations

Phase 3 Features:

- AI for automated pre-screening and intelligent workflow routing
- Predictive analytics for outcome forecasting
- Blockchain-based immutable record management

- Real-time collaborative workspaces with co-editing

Phase 4 Features:

- Continuous ML retraining pipelines from production data
- Self-service BI with drag-and-drop report builder
- AI-assisted data extraction reducing manual entry by 80%
- Open partner API marketplace for third-party extensions

Appendix H: API Endpoint Summary

Farm Management:

- POST /api/farms
- GET /api/farms/{id}
- POST /api/crop-seasons
- GET /api/crop-seasons/{id}
- GET /api/sensor-data/{farmId}

Recommendations:

- GET /api/recommendations/{seasonId}
- POST /api/pest-alerts
- GET /api/market-prices
- GET /api/weather/{farmId}

Analytics:

- GET /api/analytics/yield/{farmId}
- GET /api/analytics/input-cost
- GET /api/analytics/profitability

Auth:

- POST /api/auth/register
- POST /api/auth/login
- POST /api/auth/logout
- GET /api/users/profile

Appendix I: Component and Navigation Structure

React Components:

App.jsx:

- React Router DOM v6 with nested and protected route support
- Routes: "/" (Home), "/login" (Login), "/register" (Register), "/dashboard" (Dashboard), "/farm" (FarmDashboard)
- NavBar and Footer in root layout for all pages
- ProtectedRoute redirects unauthenticated users to login
- Global ErrorBoundary for unhandled runtime exceptions

NavBar.jsx:

- Application title: "Smart Farming and Precision Agriculture Management System"

- Navigation links: "Home" (/), "Login" (/login), "Register" (/register), "Dashboard" (/dashboard), "My Farm" (/farm)
- Logged-in user name and role from JWT token in AuthContext
- Logout clears JWT from localStorage and redirects to login
- CSS classes: nav-link, nav-active, nav-brand

Login.jsx:

- Email/username and password with show/hide toggle
- Submit "Login" → POST /api/auth/login
- JWT stored in localStorage; role saved in AuthContext on success
- Redirects to role-specific dashboard from JWT role claim
- Error: "Invalid credentials. Please check your email and password."

Register.jsx:

- Multi-step form: name, email, phone, role, domain-specific fields
- Name: alphabetic + spaces only; Phone: exactly 10 digits
- Email uniqueness verified via debounced API call
- Submit "Register" → POST /api/auth/register
- Success: "Registration successful! Please verify your email."

Dashboard.jsx:

- Role-aware cards, panels, and data tables per authenticated role
- KPI summary cards for most relevant metrics per role
- Recent activity feed with links to detail pages
- Quick-action buttons for three most-used operations
- Responsive CSS grid: single-column mobile, three-column desktop

Footer.jsx:

- "© 2024 Smart Farming and Precision Agriculture Management System. All rights reserved."
- Rendered via App.jsx layout — not imported individually
- Links to privacy policy, terms of service, and support

Document Version Control

Version	Changes	Author
1.0	Initial SRS document creation for Smart Farming and Precision Agriculture Management System	Development Team